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INCLUSION MODIFICATION AND CORROSION RESISTANCE OPTIMIZATION OF 304 STAINLESS STEEL CONTAINING CERIUM

Xiaojun Xi, Maolin Ye, Shufeng Yang✉, Jingshe Li

School of Metallurgical and Ecological Engineering, University of Science and Technology Beijing, Beijing, 100083, China

✉ Corresponding author, E-mail: yangshufeng@ustb.edu.cn

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Abstract: The effect of inclusion modification on pitting corrosion of 304 stainless steel with different cerium contents was studied by accelerated immersion test, and electrochemical measurement. The results show that in the absence of cerium, MnS and composite oxide are the dominating inclusions in stainless steel, and the average size reaches 8.6 μm, while the self-corrosion potential of steel is only -348.52 mV. By adding the cerium, the inclusions were gradually modified into spherical inclusions containing cerium, the average size is reduced, and the corrosion resistance of the steel is improved. When the content of cerium reaches 0.012 %, the MnS inclusions are all modified into spherical inclusions containing cerium, and the self-corrosion potential of steel reaches -311.25 mV, which has the best corrosion resistance. When the content of cerium increases above 0.012 %, the shape of the inclusions becomes irregular, and the size increases, resulting in a decrease in corrosion resistance.

1 Introduction

Stainless steel is widely used in chemical, petrochemical and medical industries due to its good corrosion resistance. However, stainless steel is prone to pitting corrosion in the chlorine-containing environments. Pitting corrosion is among the most dangerous modes of damage to stainless steel, not only for the difficulty in early detection as localized penetration, but also because it often serves as the precursor to crack initiation in stress corrosion and corrosion fatigue [1].

Manganese sulphide (MnS) inclusions in steel are known to be often precursor sites for pitting corrosion, which directly affects the initiation and expansion of pitting corrosion [2]. To effectively remove and control the MnS inclusions generated in the process of steelmaking and continuous casting, inclusion removal technologies such as gas stirring [3], electromagnetic purification [4], slag washing [5], and filter [6] have been developed. In addition, the effect of MnS inclusion on the properties of steel can be reduced by adding modifier to the liquid steel. At present, the main inclusions modification technologies include Ca treatment, Mg treatment, calcium-magnesium treatment, and rare earth treatment. Zhang et al. [7] indicated that Ca treatment can modify the MnS inclusions. However, if Ca reacts with sulfur to form CaS with a higher melting point than MnS, it is easy to cause nozzle clogging; if Ca reacts with oxygen to form CaO, the sulfide in steel is not completely modified, which deteriorates the

properties of steel [8]. Mg treatment can produce composite inclusions of MgS-MnS-MgO and MgS-MgO with small size, which greatly reduces the damage of the properties of steel [9]. However, the addition of excessive Mg increases the melting point of the inclusions, which reduces the castability of liquid steel [10]. The denaturation effect of calcium-magnesium composite treatment on the sulfide in steel is better than that of the single Mg treatment and Ca treatment. However, the calcium-magnesium composite treatment must have an appropriate proportion of Ca and Mg, otherwise the oxide inclusions such as CaO with high melting point are generated in steel, causing a decrease in the denaturation effect on the sulfide [11].

Cerium (Ce) has a strong affinity towards O and S, and react with these elements to form spherical or ellipsoid Ce-containing inclusions, which can significantly modify the sulfide inclusions in steel [12]. In addition, the compound formed by the reaction of Ce with S and O is easily removed in the molten steel [13]. Therefore, the Ce-containing steel can greatly reduce the corrosion tendency of the matrix induced by the MnS inclusion. In this paper, different amounts of Ce were added into stainless steel through laboratory high temperature experiments to obtain Ce-containing stainless steel. And the effects of Ce content on corrosion resistance of stainless steel were studied by means of scanning analysis, accelerated immersion test, and electrochemical experiments.

2 Experimental Design

2.1 Material Preparation

The commercial 304 stainless steel was provided by the Taiyuan Iron & Steel Co., Ltd (Shanxi Provence, China), and its chemical composition is listed in Table I. To obtain cerium-containing stainless steel, about 800 g of commercial 304 stainless steel were melted in a high-temperature electric pipe with a $MoSi_2$ heating element (Tianjin Taisite Instrument Co., Ltd, China). The furnace atmosphere was kept low in oxygen content on a constant inflow of purified argon gas, 400 $L \cdot h^{-1}$, from the bottom of the furnace. After the molten steel temperature reached 1873 K (1600°C) and kept for 30 min at this temperature, a certain amount of cerium warped in pure iron foil were added to the molten steel. A molybdenum rod was used to stir the molten steel. After the stirring, the molten steel was maintained for 30 min, and then the solidification samples could be obtained by water quenching. The quenching process was performed in water at room temperature, i.e., 293 K (20°C), and the cooling rate was about 45 $^{o}C \cdot s^{-1}$. During this study, four different mass fractions of cerium, 0, 0.006, 0.012, and 0.018 wt.%, were evaluated during the laboratory experiments.

Table I. Chemical Composition of 304 Stainless Steel (wt.%)

Element	C	Si	Mn	P	S	Cr	Ni	Fe
Content	0.060	0.480	1.380	0.020	0.002	18.370	7.210	bal.

2.2 Experimental Methods

2.2.1 Inclusion Detection

In order to clarify the inclusions in steel samples, 10×10×10 mm^3 cubic samples were cut from sample steel, then ground with 2000 grit silicon carbide paper and polished by 0.5 μm diamond paste. After polishing, about 100 typical inclusions

selected randomly from each steel sample were examined with scanning electron microscope (SEM) and energy-dispersive spectrometry (EDS) (model ProX) to determine the morphology, size and type of inclusion.

2.2.2 Electrochemical Measurements

To reveal the pitting tendency of test steels with different Ce content, 10×10×5 mm^3 cubic samples were cut from sample steel, and the exposed area (test surface) was 10×10 mm^2. A copper wire was attached to the rear side of each specimen and mounted in an epoxy resin, and the test surface was ground with 2000 grit silicon carbide paper and polished by 0.5 μm diamond paste. After polishing, the potentiodynamic polarization tests were conducted with VersaSTATMC electrochemical workstation in a three-electrode cell system, where the specimen was used as the working electrode, a platinum plate was used as the counter electrode, and a saturated calomel electrode (SCE) was used as the reference electrode. The potentiodynamic polarization measurements were measured at a scanning rate of 0.5 mV/s in 3.5wt.% NaCl solution at 298 K (25 ^oC), and the scanning range was ± 0.5 V (SCE) relative to the open circuit potential.

2.2.3 Immersion Corrosion Tests

To clarify the correlation between inclusions in test steels with different Ce contents and pitting corrosion, test specimens were immersed in the corrosion solution for which comprised 350 ml deionized water with 69.9 g $FeCl_3 \cdot 6H_2O$ and 20 ml HCl (36~38 wt.%). Each specimen was sealed with epoxy resin except for the test surface to avoid formation of porosity and cracks around the surface. The immersion time is 0 s, 30 s and 90 s, respectively. After reaching the set immersion time, the test specimen was immediately taken out of the corrosion solution, rinsed, and dried. The corrosion morphology of immersion specimen was observed with SEM.

3 Results and Discussion

3.1 Effect of Ce Content on the Morphology and Type of Inclusions

Figure 1 shows the morphology and type of inclusions in test steels with different cerium contents. Without adding the cerium, the strip-like MnS inclusion and irregular shaped Al_2O_3-SiO_2-MnO composite inclusion are mainly formed in the 304 stainless steel, which is presented in Figure 1(a), and the size of the inclusions reaches more than 10 μm. After adding 0.006 wt.% of Ce, the inclusions in test steel were modified to form Ce-containing oxygen-sulfur composite inclusions ($Ce_xO_yS_z$) and insufficiently modified MnS-$Ce_xO_yS_z$ composite inclusions. The shape of the inclusions was subsphaeroidal, and the size is reduced to about 7 to 10 μm, as shown in Figure 1(b). Figure 1(c) shows the morphology of inclusions in test steel with addition of 0.012 wt.% of Ce, the inclusions are sufficiently modified to form a single distribution of Ce-containing oxide inclusions (Ce_xO_y), Ce-containing sulfide inclusions (Ce_xS_y), and $Ce_xO_yS_z$ inclusions. The shape of the inclusions is spherical, and the size is between 3 and 10 μm. There is no MnS inclusion in test steel, indicating that the addition of 0.012 wt.% of Ce can completely modify the strip-like MnS inclusions and irregular shaped Al_2O_3-SiO_2-MnO composite inclusions. After adding 0.018 wt.% of Ce, there are mainly Ce_xS_y and $Ce_xO_yS_z$ inclusions in test steel. The shape of the inclusions becomes irregular, and the size reaches more than 10 μm.

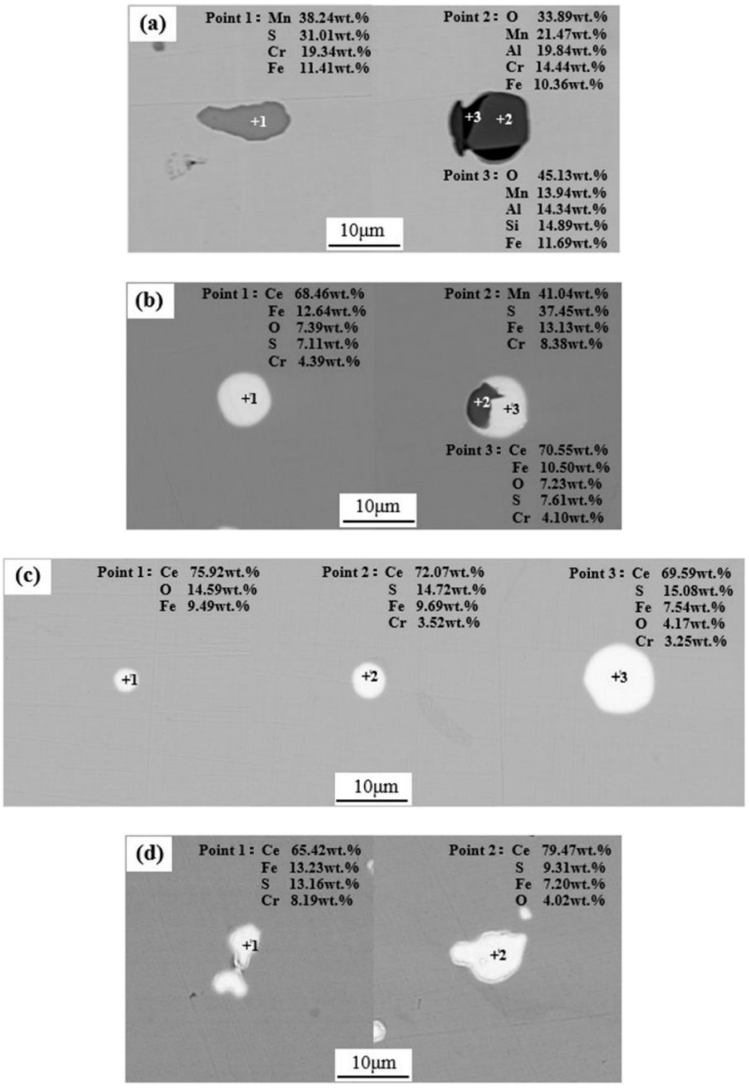


Figure 1. Morphology and composition of typical inclusions in test steels
(a) without addition of Ce; (b) with addition of 0.006 wt.% of Ce;
(c) with addition of 0.012 wt.% of Ce; (d) with addition of 0.018 wt.% of Ce

To further explain the change of the properties of inclusions in steel, 64 fields of view were randomly selected on the surface of the specimen by SEM, and the EDS of the inclusions in the field of view was analyzed. The electron microscope has a magnification of 1000 times and a field of view of 4.7 mm^2. The size of inclusions was statistically analyzed by Image-Pro Plus. Figures 2 and 3 show the proportion of various types of inclusions and the average size of inclusions in test steel, respectively. In the test steel without addition of Ce, there are mainly MnS inclusions, the proportion of MnS inclusion reaches 64 %, and there are also some irregularly shaped Al_2O_3-SiO_2-MnO composite inclusions. The average size of inclusions reaches 8.6 μm. After adding 0.006 wt.% of Ce, because the MnS inclusions in test steel are not sufficiently modified, the MnS-$Ce_xO_yS_z$ composite inclusions still exist in test steel. In addition, most of the inclusions in test steel are modified into Ce-containing inclusions, and the proportion of Ce_xO_y, Ce_xS_y and $Ce_xO_yS_z$ reaches 70 %. The average size of inclusions is reduced

to around 7.5μm. After the Ce content in test steel reached 0.012 wt.%, the inclusions were mainly Ce_xO_y, Ce_xS_y and $Ce_xO_yS_z$. Among them, the Ce_xS_y inclusions were the most, the proportion reached 47 %, and there was no MnS inclusion in test steel. The average size of inclusions in steel is only 6.8 μm. When the Ce content of test steel reaches 0.018 wt.%, the inclusions in steel are still dominated by Ce-containing inclusions, and the proportion of Ce_xO_y inclusions increases to 65 %. Compared to steel with a Ce content of 0.012 wt.%, the size of inclusions in steel has increased, with an average size of 7.1 μm.

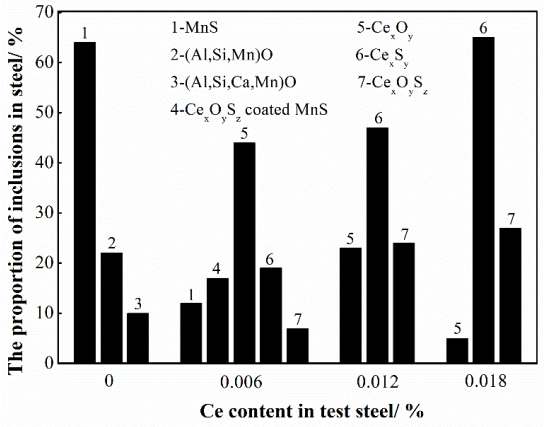


Figure 2. Proportion of inclusions in test steel with different Ce contents

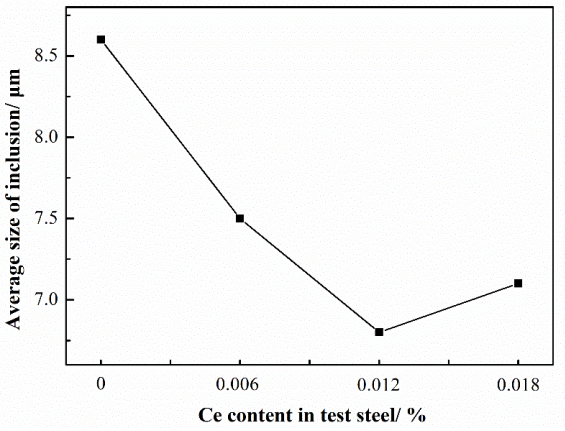


Figure 3. Average size of inclusions in test steel with different Ce contents

3.2 Effect of Ce Content on Corrosion Resistance of Test Steel

Figure 4 shows the morphology of inclusions in test steel with different Ce contents after immersion for 0 s, 30 s, and 120 s. As shown in Figure 4(a), when the test steel without addition of Ce, micro-crevices formed in the inclusion/matrix boundary at 30 s. After immersion corrosion for 2 min, the crevices in the inclusion/matrix boundary increased further, and serious pitting corrosion occurred on the surface of the steel. After adding 0.006 wt.% of Ce, slight corrosion phenomenon occurred after the test steel was immersed for 2 min, as shown in Figure 4(b). When the Ce content in test steel reaches 0.012 wt.%, there is almost no change in the surface of the steel after immersion corrosion, as shown in Figure 4(c), indicating that the test steel has excellent corrosion resistance. After adding 0.018 wt.% of Ce, pitting corrosion

occurred in the inclusion/matrix boundary after immersion for 30 s, and when the immersion time reaches 2 min, the serious corrosion phenomenon occurred on the surface of the steel, as shown in Figure 4(d).

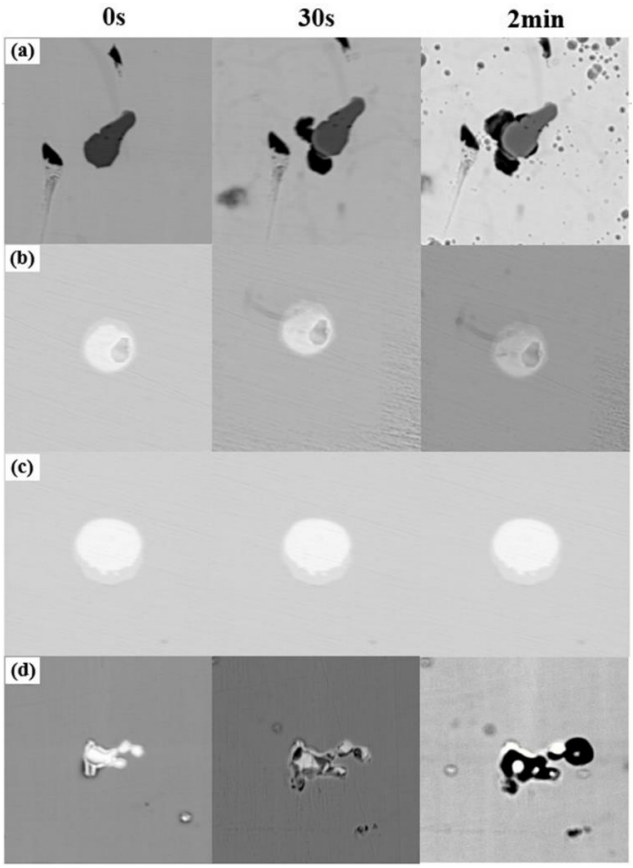


Figure 4. The morphology of inclusions in test steel after immersion at 0 s, 30 s, and 120 s
(a) without addition of Ce; (b) with addition of 0.006 wt.% of Ce;
(c) with addition of 0.012 wt.% of Ce; (d) with addition of 0.018 wt.% of Ce

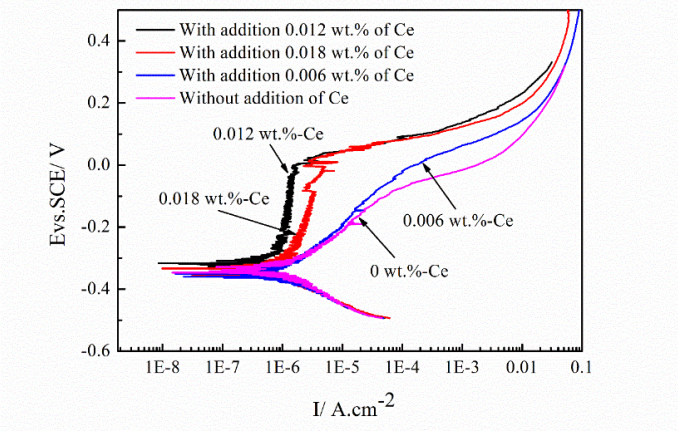


Figure 5. Electrodynamic polarization curves of test steel with different Ce contents

Figure 5 shows the electrodynamic polarization curves of test steel with different Ce content. The stainless steel without addition of Ce has the lowest self-corrosion potential, only -348.52 mV. After adding 0.006 wt.% of Ce, the corrosion potential of

test steel increased slightly, and the polarization current density was lower than that of the test steel without addition of Ce, indicating that the corrosion resistance of stainless steel is improved. When the Ce content in test steel is 0.012 wt.%, the self-corrosion potential reaches -311.25 mV, the stainless steel has excellent corrosion resistance. When the Ce content is continuously increased to 0.018 wt.%, the self-corrosion potential is lower than that of test steel with addition of 0.012 wt.% of Ce, and the corrosion current density is slightly increased, indicating that the corrosion resistance is lower than that of stainless steel containing 0.012 wt.% of Ce.

MnS inclusions are mainly present in stainless steel without addition of Ce, and its morphology, size and distribution seriously affect the properties of steel [14]. In the corrosion process, the passivation film in the inclusion/matrix boundary is weak, which is easily damaged under the corrosion of corrosive Cl^-, resulting in the activation of the matrix around the inclusions. Corrosion of the matrix produces a large amount of Fe^{3+} and Cr^{3+}, as shown in Eqs. (1) and (2). The hydrolysis reaction of Fe^{3+} and Cr^{3+} produces H^+, as shown in Eqs. (3) and (4), lead to local acidification in the inclusion/matrix boundary. The MnS inclusions are corroded to form HS^-, as shown in Eq. (5). H^+ and HS^- further corrode the matrix around the inclusions, which deteriorates the corrosion resistance of the stainless steel.

$$Fe \rightarrow Fe^{3+} + 3e^- \tag{1}$$

$$Cr \rightarrow Cr^{3+} + 3e^- \tag{2}$$

$$Fe^{3+} + 3H_2O \rightarrow Fe(OH)_3 + 3H^+ \tag{3}$$

$$Cr^{3+} + 3H_2O \rightarrow Cr(OH)_3 + 3H^+ \tag{4}$$

$$MnS + H^+ \rightarrow Mn^{2+} + HS^- \tag{5}$$

When the test steel with addition of 0.012 wt.% of Ce, the MnS inclusions were completely modified into spherical Ce-containing inclusions, which reduced the possibility of matrix corrosion induced by MnS inclusions. In addition, the shape of the Ce-containing inclusions is spherical, which has better fusion with matrix. The passivation film in the inclusion/matrix boundary is not easily damaged by the aggressive ions, which greatly reduces the probability of inclusion inducing matrix corrosion. When the stainless steel adding 0.006 wt.% of Ce, the MnS inclusions are not completely modified, and there are still existed separate MnS inclusions, the presence of MnS inclusions induce corrosion of the matrix. Therefore, the corrosion resistance is lower than that of the stainless steel containing 0.012 wt.% of Ce. After adding 0.018 wt.% of Ce, although there is no MnS inclusion in steel, the shape of the Ce-containing inclusions is irregular. The passivation film in the inclusion/matrix boundary is easily damaged by the aggressive ions, causing the matrix to be corroded.

4 Conclusions

(1) Ce can modify the MnS inclusions in steel to form Ce-containing oxide, Ce-containing sulfide and Ce-containing oxygen-sulfur composite inclusions.

(2) In corrosive environments, the corrosion products produced by the dissolution of MnS inclusion aggravate the corrosion of steel. When Ce is added, the Ce-containing inclusions greatly reduces the probability of inclusion inducing corrosion of matrix.

(3) There is an optimal addition amount of Ce in steel. under the optimal addition amount, the MnS inclusions are completely modified to form spherical Ce-containing inclusions, and inclusions have the least tendency to induce corrosion of the matrix.

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